When the Signal Breaks:

Origin-Country Elite Conflict and Vaccine Acceptance in Taiwan

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Introduction

The Puzzle: Is the COO Premium Fixed?

Vaccines are **developed** in one country but **evaluated** by publics around the world.

What We Know

- Publics prefer vaccines from technologically capable, trustworthy countries; **U.S.-developed vaccines** sit near the top (Chiang et al., 2022; Kawata & Nakabayashi, 2021; Olsen & Tuu, 2024).
- This **country-of-origin (COO) premium** extends to support for government procurement (Heinrich et al., 2024).

The Unexamined Assumption

- The origin country is assumed to send a **stable, internally consistent** signal (Frade et al., 2026).
- **Question:** When a country's own leaders publicly question its vaccines, does the premium those vaccines enjoy *abroad* erode?

The 2025 Shock and Why Taiwan

From Noise to Executive Action

- mRNA debates have circulated in the U.S. since emergency authorization.
- 2025: HHS Secretary **Robert F. Kennedy Jr.** terminated federal mRNA R&D funding on safety grounds.
- A senior executive official acting *against* the very technology in question.

A Most-Likely Case

- One of the **strongest preferences** for U.S.-developed vaccines (Chiang et al., 2022).
- By late 2021, 43.1% received an mRNA vaccine as their first COVID-19 dose (Chen, 2023).
- Vaccine politics organized around **brand choice**, shaped by partisanship and trust (Chiang & Kuo, 2025).

If U.S. elite conflict travels abroad, **Taiwan is where we should see it.**

Argument and Hypotheses

Argument: the COO premium is **politically contingent**: its credibility depends on the *coherence* of elite communication in the origin country.

Elite Cue-Taking

Elite disagreement creates uncertainty over *whom* to trust (Zaller, 1992; Hayes & Guardino, 2011).

⇒ **Whether**: H1

Negativity Bias

Negative cues outweigh reassurance (Baumeister et al., 2001; Taber & Lodge, 2006).

⇒ **Which side wins**: H2, H3

Competitive Framing

The *form* of conflict decides which consideration comes to the fore (Chong & Druckman, 2007).

⇒ **Through what**: mechanisms

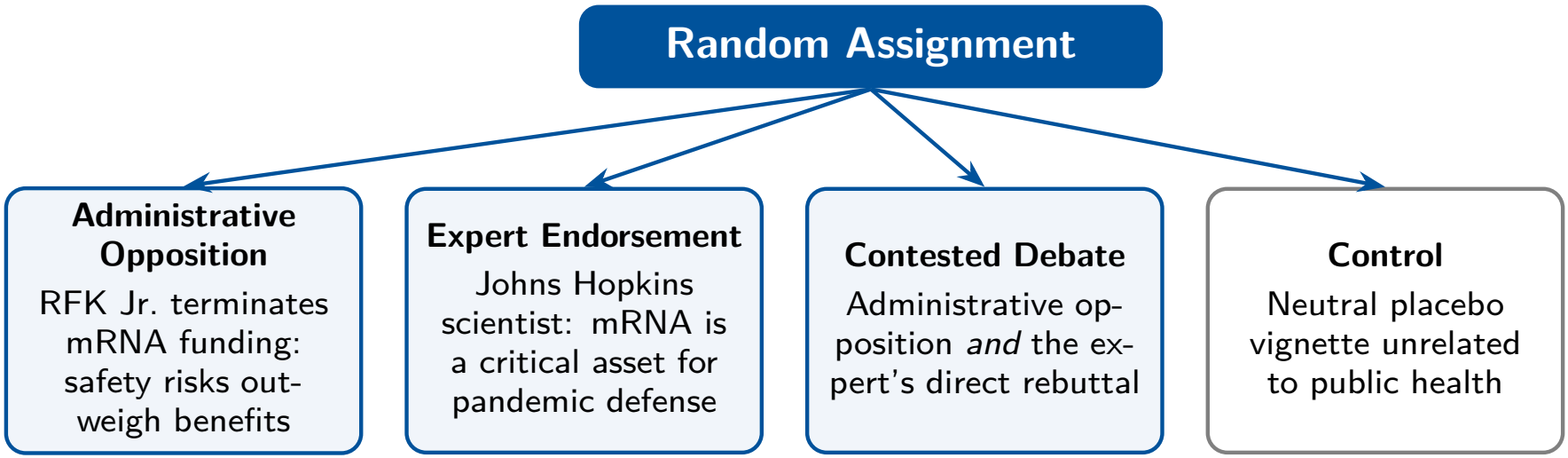
Hypotheses (pre-registered)

- **H1:** Administrative opposition *decreases* procurement support and vaccination intention.
- **H2:** Expert endorsement's positive effect is *smaller* than opposition's negative effect.
- **H3:** Opposition + endorsement together resembles *opposition alone*.
- **Mechanisms:** opposition alone → *perceived protection*; contested debate → *perceived safety*.

Research Design

Two Pre-Registered Survey Experiments

Two nationally representative waves (Oct–Nov 2025; Jan 2026), pooled $N = 4{,}147$; pre-registered on OSF (`osf.io/45crp`).



- Manipulation checks: ≈91% correctly identified the key message in every arm.

Scenario, Outcomes, and Mediators

Common Scenario (all respondents)

*Imagine that within the next month, Taiwan is threatened by a novel respiratory infectious disease. This virus is highly contagious, spreads via airborne droplets, and has a long incubation period. Once infected, patients experience severe symptoms including fever and sore throat, and may even die. Currently, the **only** available preventative measure against this disease is an **mRNA vaccine developed by a U.S. manufacturer**.*

Outcomes (1–5)

- **Policy Support:** government procurement of the vaccine.
- **Vaccination Intention:** willingness to receive it personally.

Mediators (standardized)

- **Perceived Protection:** protects against severe illness and death.
- **Perceived Safety:** no severe side effects.

Estimation: ordered probit with robust SEs; causal mediation (Imai et al., 2010) with binary outcomes. Moderators: science trust, prior mRNA vaccination.

Empirical Findings

Average Treatment Effects

	Policy Support		Vaccination Intention	
	(1)	(2)	(3)	(4)
Administrative Opposition	-0.158^{***}	-0.148^{**}	-0.132^{**}	-0.124^{**}
	(0.048)	(0.048)	(0.047)	(0.047)
Expert Endorsement	0.083	0.081	0.060	0.063
	(0.048)	(0.048)	(0.046)	(0.046)
Contested Debate	-0.160^{***}	-0.152^{**}	-0.127^{**}	-0.110^{*}
	(0.048)	(0.048)	(0.047)	(0.047)
Covariates	No	Yes	No	Yes
Observations	4,147	4,147	4,147	4,147

Table: Ordered probit coefficients (cutoffs omitted).

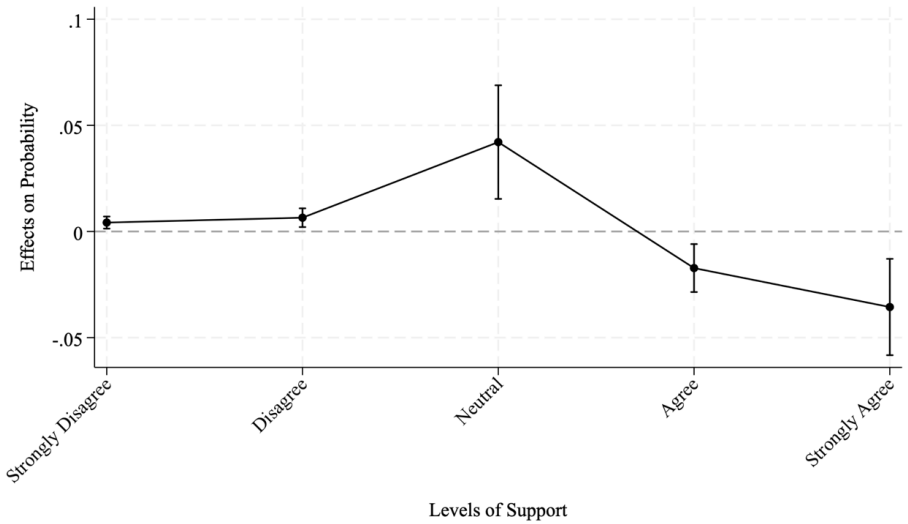
Robust SEs in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

H1 ✓ H2 ✓ H3 ✓

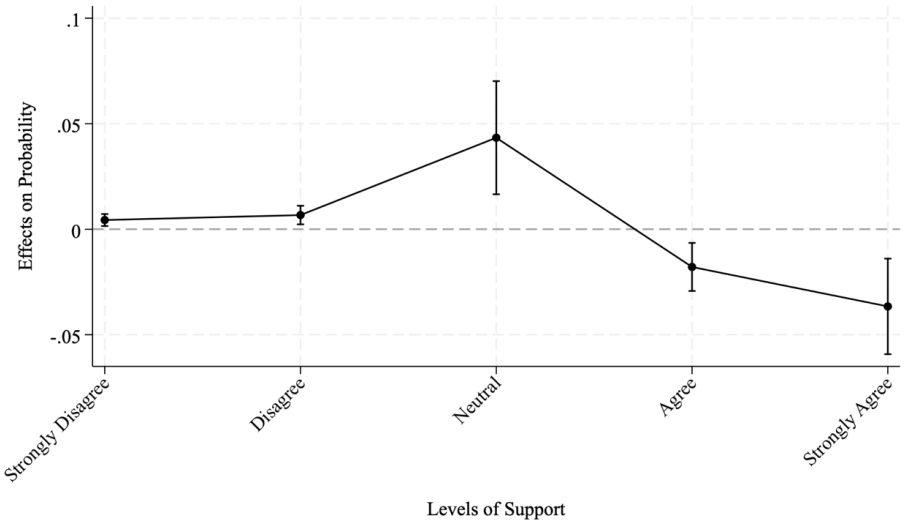
Opposition lowers both outcomes; endorsement has no significant effect; adding a rebuttal does **not** offset opposition.

Substantive Effects: Less Certainty, Not Opposition

Administrative Opposition



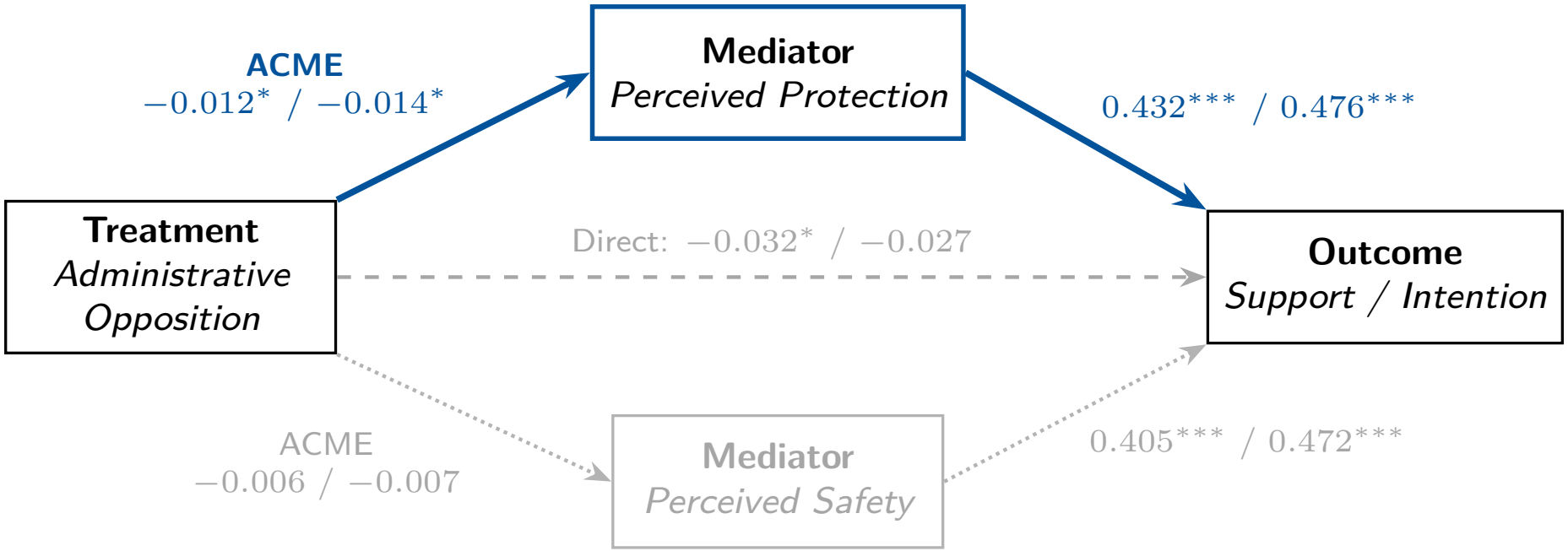
Contested Debate



- Change in predicted probability of each level of **Policy Support**.
- Mass shifts from **Agree / Strongly Agree** to the **neutral midpoint** (about $+4$ pp); "Disagree" barely moves.
- Both frames produce **virtually identical** shifts; Vaccination Intention shows the same pattern.

Mechanism I: Opposition Works Through Protection

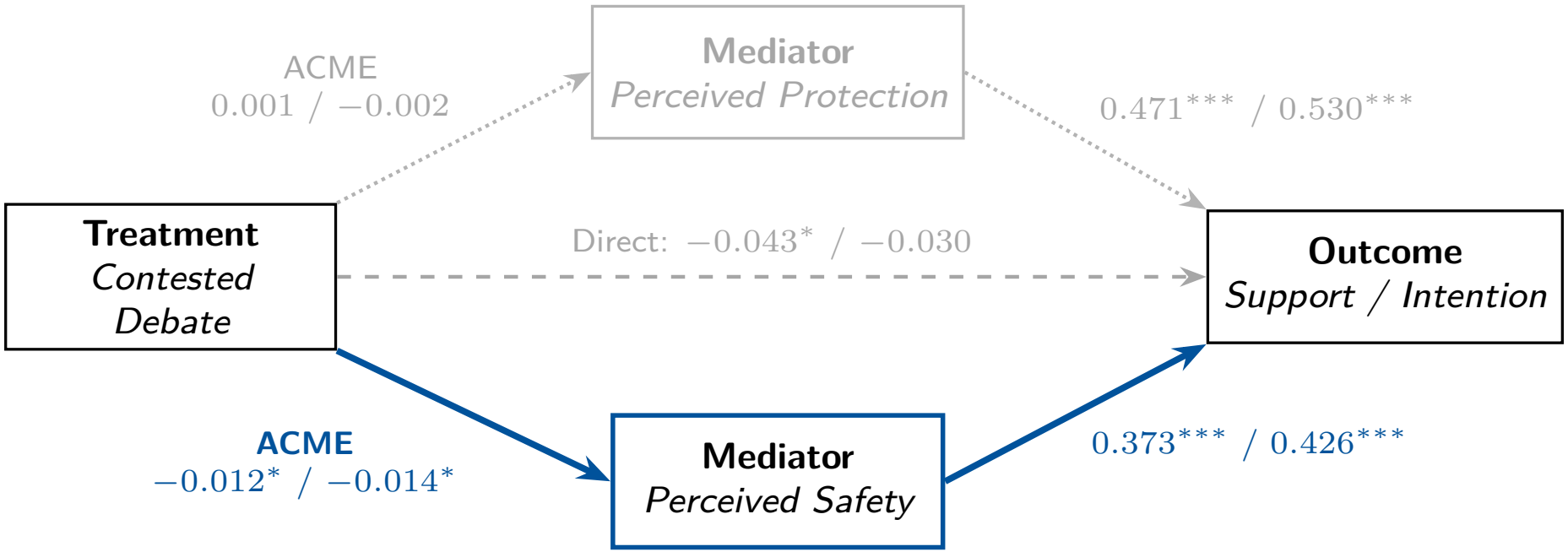
	Support	Intention
ACME via Protection	-0.012^{*} (**28%** mediated)	-0.014^{*} (**34%** mediated)
ACME via Safety	-0.006 (n.s.)	-0.007 (n.s.)



Values: Support / Intention. Probability-scale effects (binary DVs; Imai et al., 2010); mediator→outcome are probit coefficients. Each mediator estimated separately. $^{*}p < 0.05$, $^{***}p < 0.001$.

Mechanism II: Debate Works Through Safety

	Support	Intention
ACME via Safety	-0.012^{*} (**23%** mediated)	-0.014^{*} (**32%** mediated)
ACME via Protection	0.001 (n.s.)	-0.002 (n.s.)



Values: Support / Intention. Probability-scale effects (binary DVs; Imai et al., 2010); mediator→outcome are probit coefficients. Each mediator estimated separately. $^{*}p < 0.05$, $^{***}p < 0.001$.

Same Net Effect, Different Pathways

Opposition Alone → Protection

A unilateral government rejection asks: *is this vaccine effective enough to justify public investment?*

Contested Debate → Safety

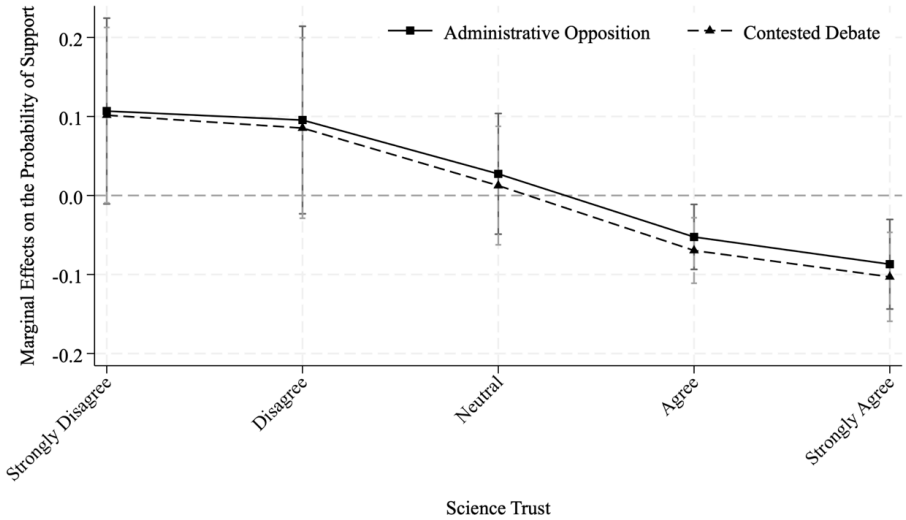
A visible clash between officials and scientists asks: *is this vaccine safe at all?*

Takeaway

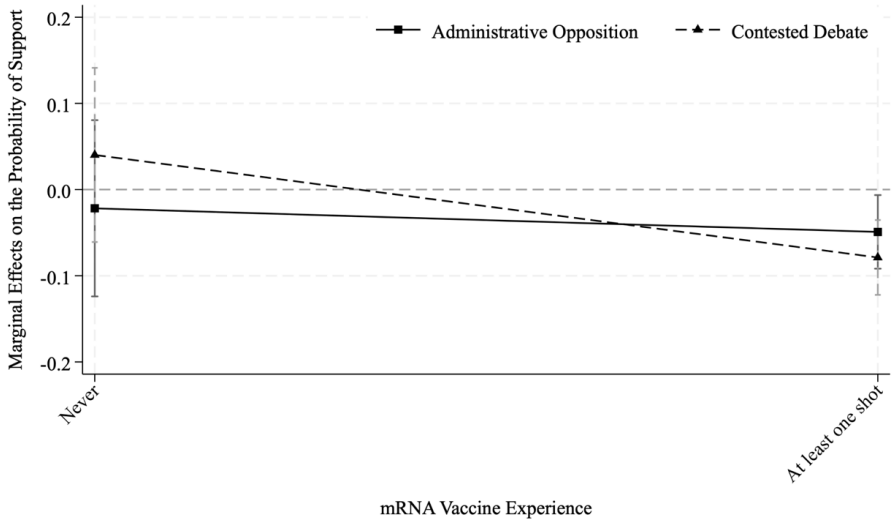
The rebuttal did not remove the negative effect, but it **changed the reasoning** behind it: the form of elite conflict determines *which dimension of risk* becomes salient.

Who Is Most Affected? (Exploratory)

Science Trust × Policy Support



Prior mRNA Experience × Policy Support



- **Higher science trust** ⇒ larger drops in policy support, while personal intention stays more resilient.
- **Prior mRNA recipients** ⇒ larger drops in both outcomes; no significant effect among the never-vaccinated.
- The **most invested** publics are the **most vulnerable** to credibility disruption.

Implications

Implications for Theory and the Literature

1. **COO effects:** the COO signal is **politically constructed**, not exogenous. First experimental evidence that a well-documented premium erodes when the origin country fights over its own technology (Verlegh & Steenkamp, 1999; Chiang et al., 2022).
2. **Negativity bias, transnationally:** opposition suppresses the premium far more than endorsement restores it; credibility is **easier to undermine than to restore** (Baumeister et al., 2001; Heinrich et al., 2024).
3. **Competitive framing and conflicting cues:** elite conflict shapes not only *whether* people update but *which dimension of risk* they weigh, extended to a cross-border setting (Chong & Druckman, 2007; Darmofal, 2005; Leeper & Slothuus, 2014).

Implications for Policy

1. **Domestic conflict is an international public health concern:** political disputes in an exporting country shape whether foreign publics accept its technologies.
2. **Match the message to the pathway:** restoring belief in *effectiveness* requires different messages than restoring confidence in *safety*.
3. **Expert voices still matter, but are not enough:** once executive opposition undermines origin-country institutions, scientific endorsement alone may not restore confidence.

Bottom Line

Elite conflict in the origin country can erode an advantage long treated as fixed, carrying **domestic disputes into vaccine attitudes abroad**.

Thank you!

Q & A

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Backup: Elite Conflict and the Fragility of COO Signals

Argument: COO premiums are **politically contingent**, not fixed — signal credibility depends on the *coherence* of elite communication in the origin country.

Elite Cue-Taking

- Citizens rely on trusted elites for technical issues (Zaller, 1992; Lupia & McCubbins, 1998).
- Disagreement ⇒ uncertainty over *whom* to trust.

Negativity Bias

- Endorsement *reinforces* existing beliefs ⇒ modest gains (Barceló et al., 2023).
- Negative information carries greater weight (Taber & Lodge, 2006).
- Opposition + endorsement ⇒ opposition should dominate.

Why It Travels Across Borders

Citizens update on foreign elites (Hayes & Guardino, 2011), and mRNA risks are **universal** across borders.

Backup: Hypotheses

H1 (Administrative Opposition Hypothesis)

Exposure to executive-branch opposition to mRNA technology in the origin country **decreases** foreign publics' support for government procurement of that vaccine and their intention to receive it.

H2 (Expert Endorsement Hypothesis)

The positive effect of scientific endorsement will be **significantly smaller** than the negative effect of executive opposition.

H3 (Contested Debate Hypothesis)

The effect of simultaneous exposure to executive opposition and expert endorsement will more closely resemble that of **executive opposition alone** than that of expert endorsement alone.

Backup: Data: Two Nationally Representative Survey Waves

Sample

- **Wave 1:** Oct 22 – Nov 2, 2025 ($n = 2{,}081$), treated as pilot.
- **Wave 2:** Jan 9 – 19, 2026 ($n = 2{,}066$), pre-registered on OSF (`osf.io/45crp`).
- Quota sampling on gender, age (20–55), and four major regions.

Why Pool?

- Identical experimental design $\Rightarrow$ pooled $N = 4{,}147$.
- Improves statistical precision and estimation stability (Baltagi, 2005; Imai et al., 2010).
- Survey wave included as a control.

Backup: Mediators and Moderators

Mediators

- **Perceived Safety:** the vaccine would *not* cause severe side effects.
- **Perceived Protection:** the vaccine would protect against severe complications and death.
- Correspond to safety- vs. efficacy-based appraisal (Chiang et al., 2022).

Pre-treatment Moderators

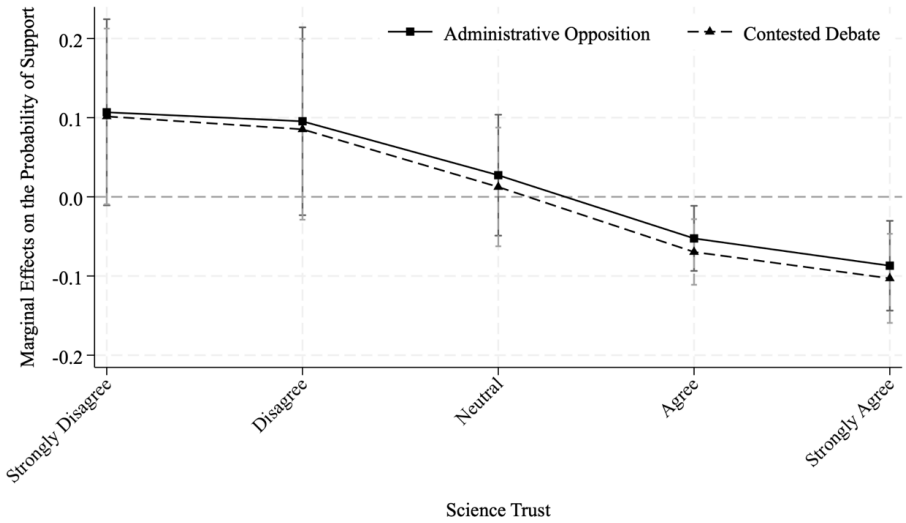
- **Science Trust:** Nadelson et al. (2014) battery; IRT (GRM) identifies ST3 as most informative ($\alpha = 2.75$).
- **Prior mRNA Experience:** $\geq$1 dose of Moderna or Pfizer-BNT (82.7%); exploratory.

Harmonizing Mediators Across Waves

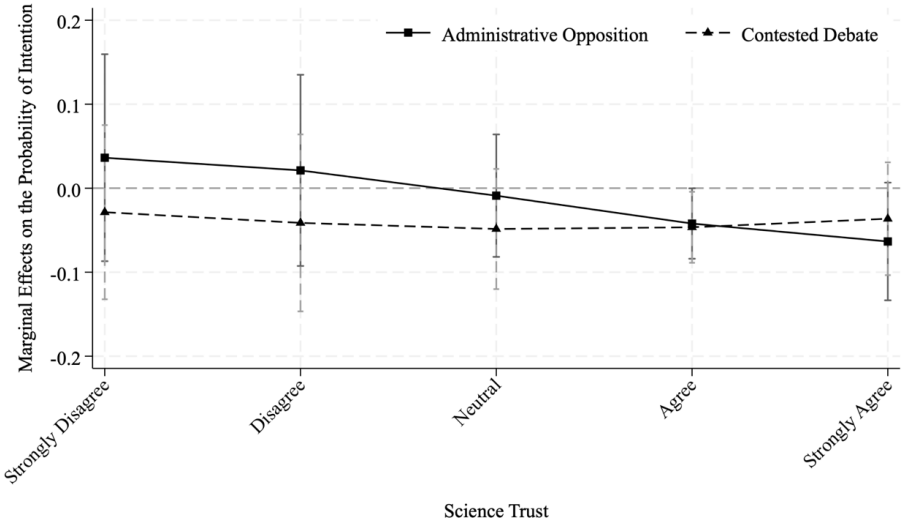
Wave 1 used 0–100 feeling thermometers; Wave 2 used 5-point Likert items $\Rightarrow$ **standardized within wave** before pooling.

Backup: Heterogeneity by Science Trust

Policy Support



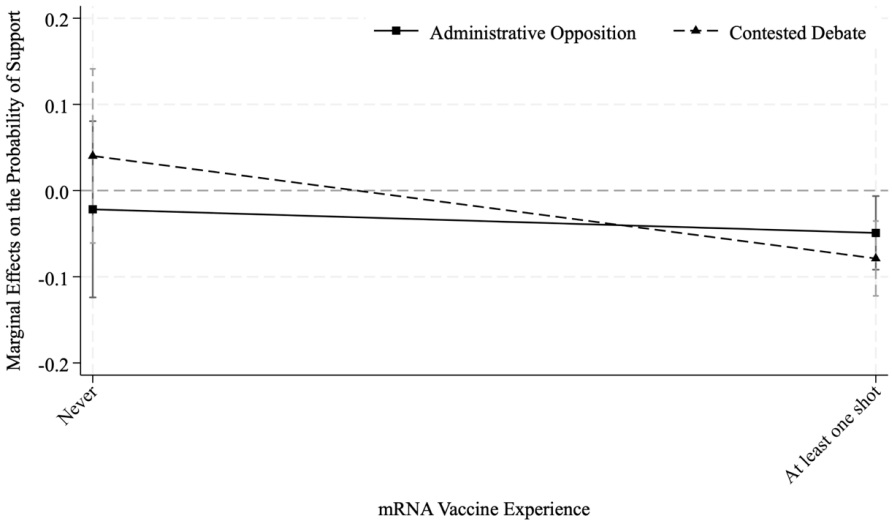
Vaccination Intention



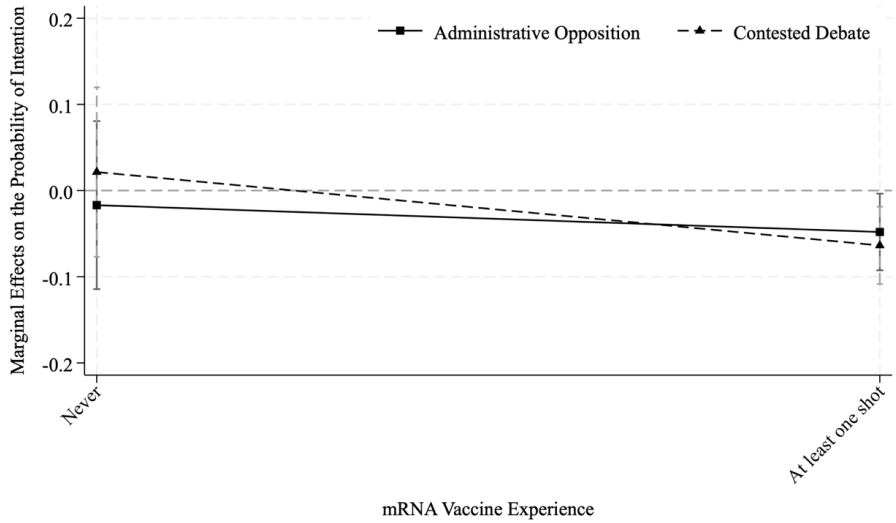
- **Policy Support:** effects grow more negative as Science Trust rises — consistent with elite cue-taking by institution-reliant citizens.
- **Vaccination Intention:** significant effects concentrate at moderate-to-high trust; the highest-trust group remains comparatively resilient.
- High-trust respondents **decouple** policy-level from behavioral-level responses.

Backup: Heterogeneity by Prior mRNA Experience

Policy Support



Vaccination Intention



- Respondents with $\geq$1 mRNA dose show **larger reductions** on both outcomes; effects among the never-vaccinated are insignificant.

Counterintuitive

Direct experience **heightens** rather than buffers susceptibility: those already committed to the technology have a greater personal stake in its credibility.

Backup: Country-of-Origin Effects in Vaccine Evaluation

The American Vaccine Premium

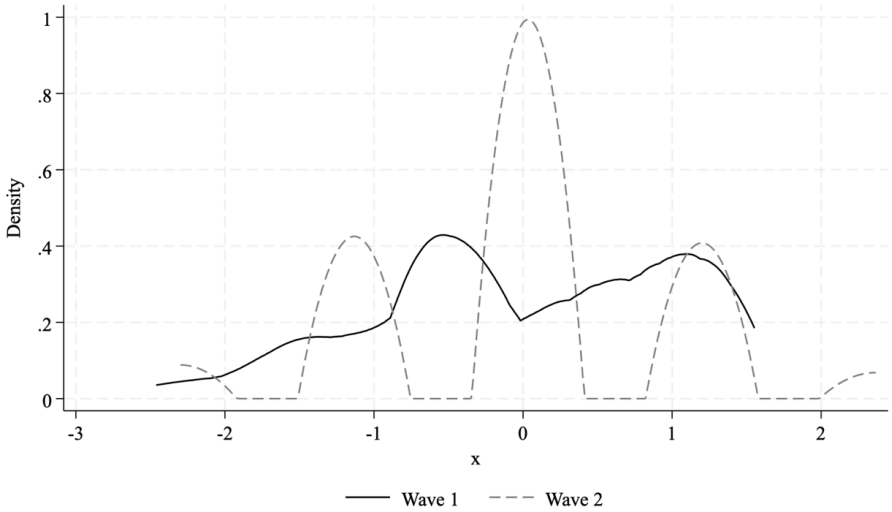
- **Japan:** U.S. over Chinese by >27 pp (Kawata & Nakabayashi, 2021).
- **Vietnam:** U.S. image raises acceptance (Olsen & Tuu, 2024).
- **Taiwan:** U.S. over Chinese vaccines, mediated by perceived efficacy and side-effect risk (Chiang et al., 2022).
- **U.S.:** COO shapes procurement support; endorsements barely neutralize it (Heinrich et al., 2024).

The Theoretical Blind Spot

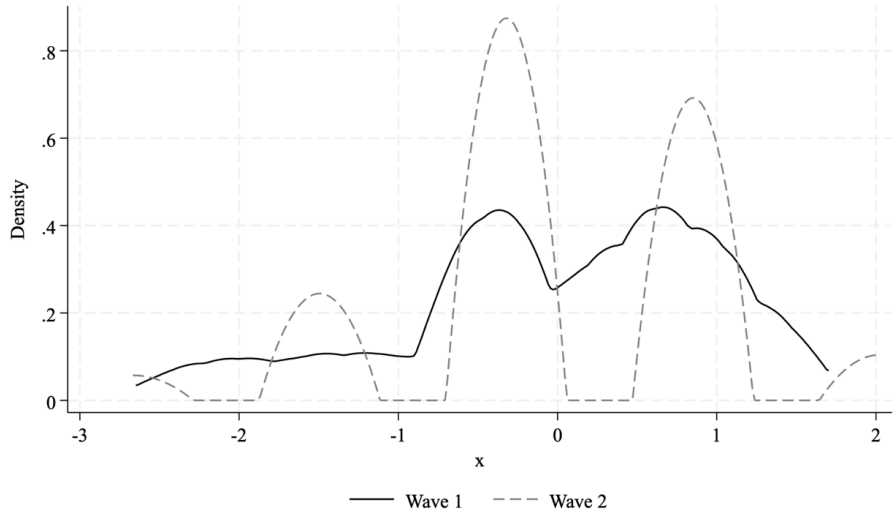
- COO as an **information heuristic** about vaccine quality.
- Evidence comes from origin countries speaking with **one voice**.
- Does the premium survive **publicly contested** elite signals?

Backup: Harmonization Check: Kernel Densities by Wave

Perceived Safety



Perceived Protection



- Standardized distributions are comparable across waves, with no meaningful differences in skewness or kurtosis.

Backup: Estimation Strategy

Average Treatment Effects

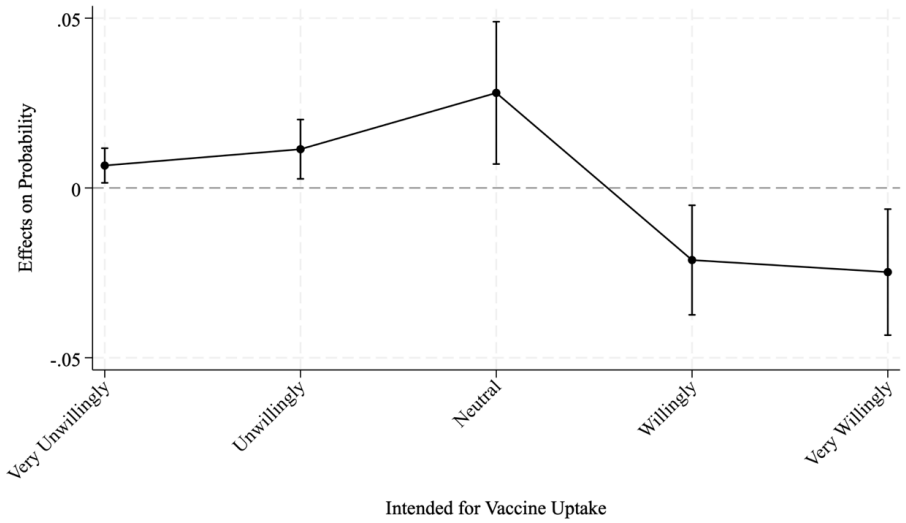
- **Ordered probit** with robust standard errors (Liao, 1994; Long, 1997).
- Two specifications per DV: treatment indicators only; plus pre-treatment covariates.
- Covariates: survey wave, prior COVID-19 and flu vaccination, age, gender, education, employment, household income, region.
- Minor finite-sample imbalances (Appendix A) $\Rightarrow$ covariate adjustment keeps estimates conservative.

Mechanisms and Heterogeneity

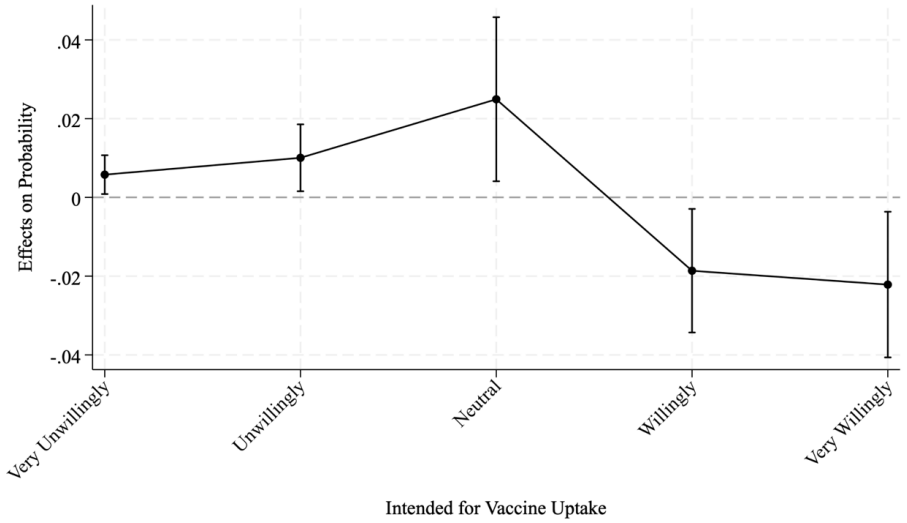
- **Causal mediation** (Imai et al., 2010): DVs dichotomized (4–5 = support / intend); estimated separately for Administrative Opposition and Contested Debate.
- **Heterogeneous effects** by Science Trust and prior mRNA experience.

Backup: Substantive Effects: Vaccination Intention

Administrative Opposition



Contested Debate



Uncertainty, Not Radicalization

Negative frames do not convert supporters into opponents; they **weaken the certainty** behind previously favorable evaluations. Scientific rebuttal does not recover the lost mass at the top of the scale.

Backup: Causal Mediation: Administrative Opposition

	Policy Support		Vaccination Intention	
Mediator	Safety	Protection	Safety	Protection
ACM Effect	-0.006	-0.012^{*}	-0.007	-0.014^{*}
Direct Effect	-0.039^{*}	-0.032^{*}	-0.035^{+}	-0.027
Total Effect	-0.045^{+}	-0.044^{+}	-0.042	-0.040
% Mediated	—	**27.87%**	—	**33.71%**
Observations	2,073	2,073	2,073	2,073

Table: Causal mediation effects, Administrative Opposition vs. control.

$^{+}p < 0.1$, $^{*}p < 0.05$. Binary DVs; covariates included.

- No significant indirect effect through **Perceived Safety**.
- Significant indirect effect through **Perceived Protection** (28–34% mediated).
- Unilateral opposition lowers beliefs that the vaccine *protects*, rather than amplifying side-effect concerns.

Backup: Causal Mediation: Contested Debate

	Policy Support		Vaccination Intention	
Mediator	Safety	Protection	Safety	Protection
ACM Effect	-0.012^{*}	0.001	-0.014^{*}	-0.002
Direct Effect	-0.043^{*}	-0.056^{***}	-0.030	-0.044^{*}
Total Effect	-0.055^{*}	-0.055^{*}	-0.044^{+}	-0.043
% Mediated	**22.69%**	—	**32.32%**	—
Observations	2,074	2,074	2,074	2,074

Table: Causal mediation effects, Contested Debate vs. control.

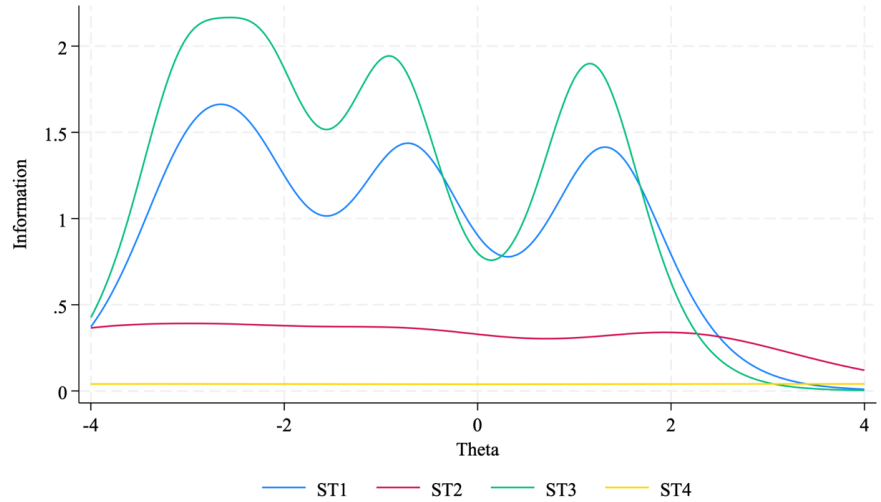
$^{+}p < 0.1$, $^{*}p < 0.05$, $^{***}p < 0.001$. Binary DVs; covariates included.

- The pattern **reverses**: significant indirect effect through **Perceived Safety** (23–32% mediated).
- No meaningful indirect effect through **Perceived Protection**.

Appendix: Measuring Science Trust (IRT)

Item	Wording
ST1	I trust the work of scientists to make life better for people.
ST2	We cannot trust science because it moves too slowly.†
ST3	I trust scientists can find solutions to our major technological problems.
ST4	Scientists ignore evidence that contradicts their work.†

† Reverse coded. Adapted from Nadelson et al. (2014).



Item information curves (Graded Response Model)

Item	α	β_1	β_2	β_3	β_4
ST1	2.37	−2.92	−2.43	−0.71	1.33
ST2	1.14	−4.04	−2.59	−0.71	2.10
ST3	**2.75**	−3.00	−2.45	−0.88	1.17
ST4	0.38	−9.88	−3.21	3.39	9.88

- ST3 has the highest discrimination and peaks highest across the latent continuum ⇒ used to represent Science Trust in marginal-effect plots.

Appendix: Balance Across Treatment Groups

	Admin. Opp.	Expert End.	Contested	Control mean
Age	0.249	−0.469	−0.158	38.941
Female	0.019	−0.003	0.011	0.491
College	−0.014	−0.009	0.005	0.815
Employed	−0.020	−0.015	−0.031*	0.884
Wave 2	−0.000	−0.001	−0.000	0.499
Covid Vax.	0.018	0.004	−0.020	0.939
Flu Shot	−0.052*	−0.022	−0.029	0.539
Family income 0	0.005	−0.008	0.011	0.054
Family income 1	0.019	0.014	0.034*	0.135
Family income 2	−0.019	−0.019	−0.042*	0.335
Family income 3	−0.014	0.016	−0.037	0.336
Family income 4	0.009	−0.003	0.035*	0.141
North	0.042	0.022	0.014	0.450
Central	−0.037*	−0.044*	−0.020	0.270
South	−0.007	0.024	0.011	0.257
East	0.003	−0.002	−0.004	0.023

Table: Difference-in-means relative to control. $^*p < 0.05$.

Minor imbalances motivate the covariate-adjusted specifications.